



Mohammed Omar Khan

SYSTEMS & PLATFORM ENGINEER - MLOPS

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Summary

Systems engineer with experience across infrastructure, data engineering and ML, currently running production platforms inside an air-gapped UAE government environment: OpenShift, Nutanix, Linux/Solaris servers and a centralised Elasticsearch observability stack. Outside work, I build and operate a self-hosted Kubernetes platform that serves open-weight LLMs through vLLM behind an OpenAI-compatible API, with Prometheus/Grafana monitoring, persistent storage and GitOps-style tooling. Looking to move into MLOps, taking models from experimentation to reliable, observable production systems.

Experience

Midis Group

Abu Dhabi, UAE

SYSTEMS ENGINEER - DEPLOYED ONSITE TO A UAE GOVERNMENT AGENCY

March 2024 - Present

- Placed onsite as a technical point of contact at a UAE government agency inside a strictly network-isolated (**air-gapped**) environment via a technology partner where I administer the customer environment and translate operational requirements into deployed solutions.
- Designed, shipped and operate a centralised **Elasticsearch + Kibana observability platform** used by stakeholders.
- Used N8N to automate workflows to improve efficiency in the workplace by at least 20%.
- Administer a **Nutanix and Red Hat OpenShift** environment, along with mixed Linux and Solaris servers, and a local Elastic Cloud Enterprise environment, with tangential support to GPU-equipped servers
- Author Python and Bash automation for operational data and reporting workflows, and use Ansible for infrastructure automation.
- Coordinate with vendor teams and debug issues across OS, application, network and platform layers

MDS TSS (Midis Group)

UAE

BIG DATA INTERN

May 2023 - March 2024

- Completed Hadoop training and contributed to customer big-data engagements (cluster deployment exercises, pipeline tuning).
- Authored Python automation for customer data-mining and analysis workflows, where I produced operational reports on ticket throughput and SLA performance for stakeholders.

Heriot-Watt University

Dubai, UAE

TEACHING ASSISTANT

Sept. 2022 - May 2023

- Mentored 50+ Computer Science students in Python, machine-learning algorithms, data visualisation, Jupyter and ML tooling through regular lab sessions.
- Assisted faculty across undergraduate Computer Science modules, including AI/ML-focused courses.

Skills

Agentic AI & LLMs

LangGraph, LangChain, tool-calling / Hermes parser, MCP, RAG, Chroma vector store, local embeddings, Qwen / DeepSeek / LLaMA

LLM Serving

vLLM, TGI, Ollama, quantization, Hugging Face ecosystem, MLflow

ML Frameworks

PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, SVMs, NLP, hyperparameter tuning

Software & APIs

Python, Flask, FastAPI, REST APIs, microservices, PyPi packaging, Git

Infra, DevOps & Deployment

Kubernetes, Docker, ProxMox, Terraform, Ansible, **OpenShift Pipelines**, Nutanix, Red Hat OpenShift; Elastic Cloud Enterprise (ECE)

Observability & Data

Elasticsearch, Kibana, Grafana, Pandas, NumPy, SQL, ETL pipelines, Hadoop

Operating Systems

Linux (RHEL, Rocky, Ubuntu, Debian), Solaris, bash scripting

Programming

Python, SQL, Bash, C, C++, Java, JavaScript

Languages

English, Hindi, Urdu

Projects

Local GenAI Stack - LLM Serving, RAG, and Agentic Orchestration

vLLM, TGI, OLLAMA, KUBERNETES, LANGCHAIN, LANGGRAPH, CHROMA

- Deployed open-weight LLMs from the **Qwen, DeepSeek and LLaMA** families fully offline on a self-managed Kubernetes cluster, benchmarking **vLLM, TGI (Hugging Face Text Generation Inference) and Ollama** as inference runtimes side-by-side.
- Configured **tool-calling for agentic workflows** via vLLM's auto tool-choice mode with a Hermes-style parser, exposing an OpenAI-compatible endpoint that agent frameworks can drive natively.
- Built agent orchestration flows with **LangGraph** on top of the served models — stateful multi-step workflows with tool selection, conditional edges, and handoff patterns; explored persistent memory strategies for context-window management.
- Ran **quantized variants (GGUF, GPTQ, AWQ)** to fit larger models into constrained VRAM; measured throughput / latency tradeoffs across quantization levels and runtimes to pick the right combination for each use-case.
- Built the **RAG** pipeline in LangChain over an Obsidian Markdown knowledge base: chunking on heading boundaries, embedding with a local model (nomic-embed-text), storing vectors in Chroma; tuned retrieval top-K and prompt construction to balance relevance against context-window budget.

Self-Hosted Kubernetes Platform

KUBERNETES, PROXMOX, HELM, PROMETHEUS, GRAFANA, LOKI, TAILSCALE

- Built and operate a 3-node kubeadm Kubernetes cluster on Proxmox and bare-metal nodes, running services such as file sync, document management, self-hosted AI model stack and Forgejo, a self-hosted Git.
- Designed the storage layer with dynamic PV provisioning through CSIs, CloudNativePG for stateful Postgres workloads, and backups.
- Implemented cluster networking and access control, along with Tailscale to manage remote connectivity to the cluster applications
- Deployed full-stack observability with kube-prometheus-stack, Grafana, Loki and Alloy for metrics and logs across cluster and application workloads.

Advertisement Detection (Final-Year Capstone)

PYTORCH, TENSORFLOW, SCIKIT-LEARN, NLP

- Personally implemented and trained classifiers on a 1M+ row URL dataset assembled from multiple sources
- Engineered NLP-based features from URL structure to detect ad links.
- Compared neural networks, XGBoost, SVMs, LightGBM and Random Forests to find the best-performing model and reached 80% classification accuracy.

Sentiment Analysis - Amazon Reviews (227k) and Twitter (10k+)

TENSORFLOW, SCIKIT-LEARN, NLP, PANDAS

- Two end-to-end NLP pipelines: ETL, text preprocessing, feature engineering, model training, evaluation and visualisation.
- Achieved 90% classification accuracy on Amazon and 92% on Twitter by combining neural networks with classical methods (XGBoost, SVMs, Random Forests) and hyperparameter tuning.

vscan - Open-source Python security scanner

PYTHON, REST APIs, PYPI, GITHUB CI/CD PIPELINES

- Project lead on a Python-based virus scanner that uses different APIs to scan files and websites for malicious content.
- Built a GitHub Actions CI/CD pipeline that runs tests on every pull request and automatically builds and publishes versioned releases to PyPi on tagged commits.

Education

Heriot-Watt University

B.Sc.(HONS) COMPUTER SCIENCE WITH ARTIFICIAL INTELLIGENCE

Dubai, UAE

Sep. 2020 - June 2023

- Upper Second Class Honours (2:1) - Awarded the Deputy Principal's Award for academic excellence.
- AI-focused coursework: neural networks, supervised and unsupervised machine learning, applied text analysis (NLP), and computer vision foundations.
- Final-year capstone in ML-based advertisement detection.

Certificates

- 2024 **Data Science A-Z**, SuperDataScience
- 2023 **Azure Data Fundamentals**, Microsoft
- 2023 **Career Essentials in Data Analysis**, Microsoft
- 2021 **Certified Junior Associate – Java**, Oracle